QNN AND ERROR CORRECTION

A neural-network decoder — Steane-code syndrome decoding and model comparison

Problem statement

“QNN and error correction — A neural-network decoder. Train a quantum or classical network to decode Steane syndromes, and compare with the lookup table.”

1. Objective and evaluation target

This project studies whether learned decoders can map noisy syndrome observations to reference correction labels for the seven-qubit Steane code. Four approaches are compared: a deterministic lookup-table baseline, a feedforward multilayer perceptron (MLP), a recurrent GRU, and a hybrid quantum neural network (QNN). The supplied metrics evaluate **correction-label accuracy** and separate X/Z component accuracies—not end-to-end logical error rate.

2. Experimental setup

The supplied project artifacts describe 6,000 training, 1,200 validation, and 1,500 test examples. Each trajectory contains three rounds of six-bit syndrome observations, with data-noise parameter *p* and syndrome measurement-flip parameter *q*. The lookup and MLP use the latest observed syndrome; the GRU uses all three rounds. The QNN uses a local exact StatevectorEstimator simulator; no quantum hardware was used.

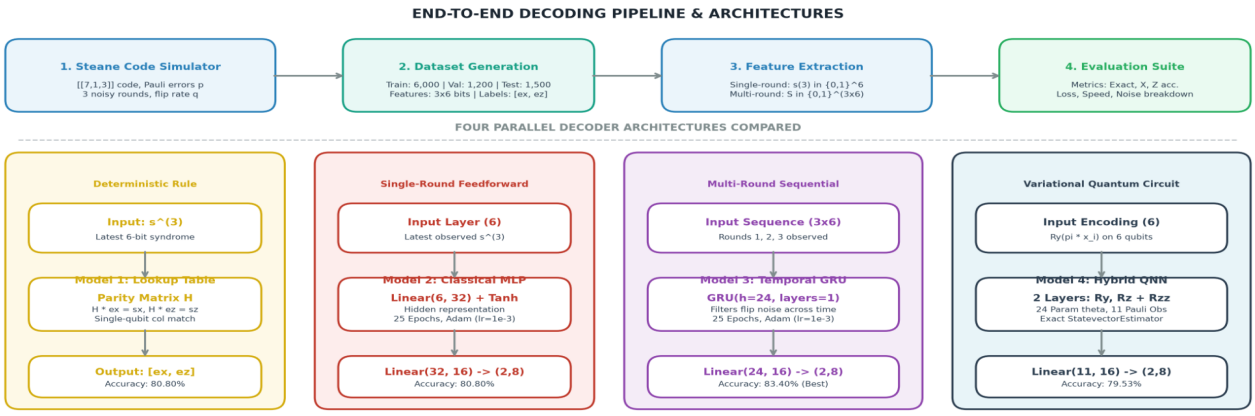


Figure 1. Overall project pipeline and four decoder branches (reproduced from the supplied architecture diagram).

3. Decoder architectures

All four methods share the same broad task—predict the pair of reference correction labels—but they do not use identical input information. This matters when interpreting the comparison.

Model 1 — Lookup table
Input: latest observed 6-bit syndrome
Apply parity-check relationships / deterministic syndrome-to-correction rule
Output: reference pair [e■, e_z]

Model 2 — Classical MLP
Input: latest observed syndrome (6 features)
Linear(6, 32) + tanh
Linear(32, 16) → output logits for correction labels

Model 3 — Temporal GRU
Input: 3 rounds × 6 syndrome bits
GRU: hidden size 24, one layer
Linear(24, 16) → output logits

Model 4 — Hybrid QNN
Input encoding: 6 qubits, R■(π·x■)
Trainable variational layers: R■ / R■ rotations and ZZ entanglers
Expectation-value features → classical linear head → correction logits

Architecture interpretation

The lookup table is deterministic and provides a reference baseline. The MLP learns a nonlinear mapping from one six-bit observation. The GRU can use temporal information across three rounds, which can help distinguish persistent syndrome structure from measurement flips. The QNN encodes syndrome bits into a six-qubit circuit and combines measured expectation values with a classical output layer. Its parameterized circuit is a hybrid model, not a purely quantum end-to-end system.

4. Results and quantitative comparison

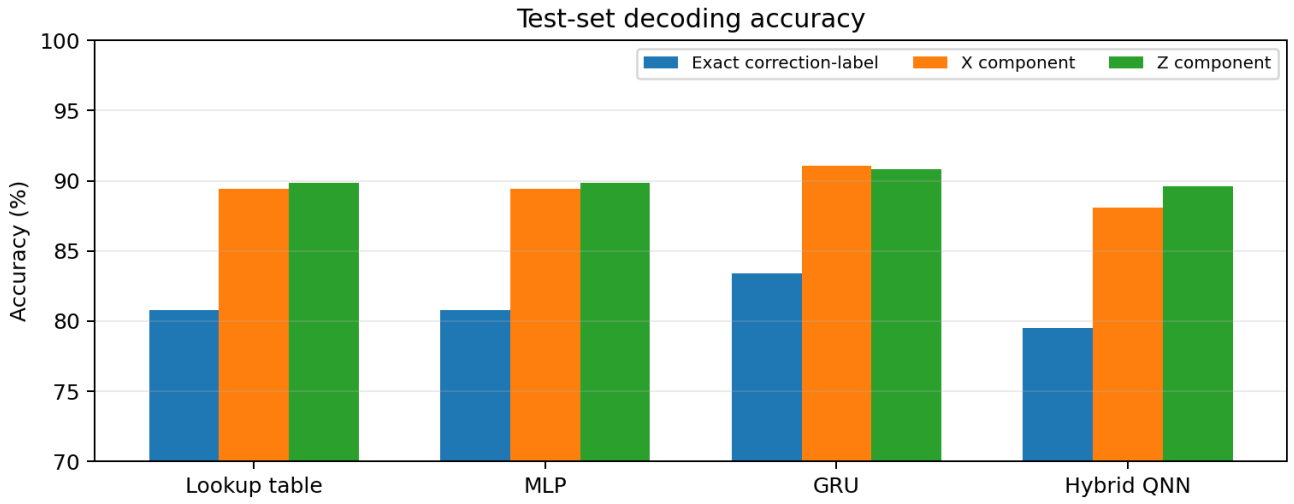


Figure 2. Test-set accuracy from the supplied metrics JSON. Values are percentages; exact correction-label accuracy requires both components to match.

Decoder	Exact label	X accuracy	Z accuracy	Cross-entropy	Training time
Lookup table	80.80%	89.40%	89.87%	—	—
MLP	80.80%	89.40%	89.87%	0.488	2.0 s
GRU	83.40%	91.07%	90.80%	0.341	3.5 s
Hybrid QNN	79.53%	88.07%	89.60%	0.424	147.7 min

Main findings

GRU ranks first at 83.40% exact correction-label accuracy, 2.60 percentage points above lookup/MLP (80.80%) and 3.87 points above QNN (79.53%). GRU also has the highest X-component accuracy (91.07%) and Z-component accuracy (90.80%). Its advantage is consistent with access to three rounds of syndrome history, while lookup and MLP use only the latest round.

The MLP matches the lookup baseline on the supplied test set (80.80% exact). This does not show that the MLP is useless: it shows no measured accuracy gain under this particular feature/label setup. The QNN reaches 79.53% exact accuracy, below lookup by 1.27 percentage points, while requiring far more training time in the supplied run.

Cross-entropy is available for MLP (0.488), GRU (0.341), and QNN (0.424); it is not defined for the deterministic lookup table. Lower loss indicates better probabilistic fit under the recorded loss, but accuracy is the more direct metric for exact correction-label decoding.

5. Noise sensitivity, quantum circuit, and limitations

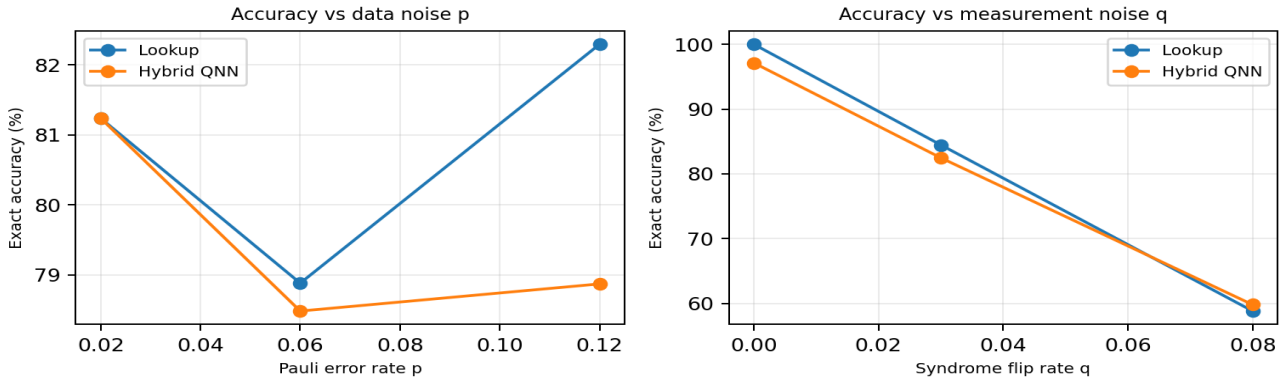


Figure 3. Lookup and QNN exact accuracy by noise parameter, from the supplied noise-breakdown CSV. Interpret each point as the reported test subset for that parameter value.

Noise trend

Both decoders degrade as measurement-flip noise *q* increases: lookup falls from 100.0% at q=0.00 to 58.82% at q=0.08, while QNN falls from 97.13% to 59.80%. At q=0.03, lookup is 84.46% and QNN is 82.47%. Across data-noise values p, the pattern is not strictly monotonic in the supplied groups; therefore, avoid claiming a smooth or universal trend from only three sampled values. Subset composition and finite sample sizes can affect the points.

Quantum circuit used by the QNN

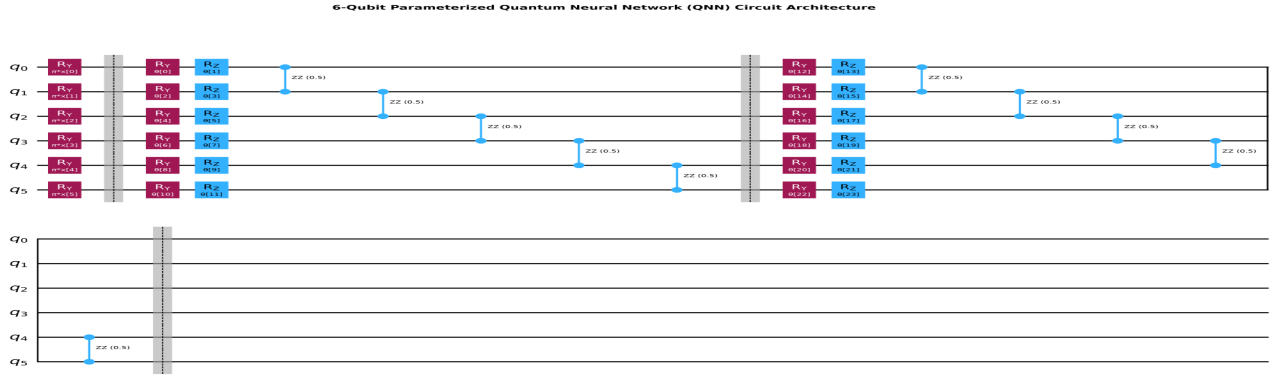


Figure 4. Supplied six-qubit parameterized circuit diagram. The model encodes six input bits using R■ rotations, applies trainable R■/R■ rotations and ZZ entangling interactions in two variational blocks, then uses expectation values as features for a classical output head.

Efficiency and limitations

The supplied metrics record QNN training time of 147.7 minutes and test-set inference time of 24.7 seconds. In contrast, MLP and GRU training times are about 2.02 s and 3.52 s in the same recorded run. These timings depend on hardware, software versions, simulator settings, and batching; they should not be treated as hardware-independent benchmarks.

Important scope limitation: the supplied labels are reference single-qubit correction labels derived from the final true syndrome. The project does not implement a full recovery/logical-observable simulation, so these numbers are not logical-error rates and do not prove fault-tolerant quantum error correction. No quantum hardware was used; QNN results come from exact statevector simulation. The model inputs differ (latest round for lookup/MLP versus three rounds for GRU), so the comparison is useful but not a fully controlled architecture-only comparison.

Conclusion

For this dataset and label definition, the GRU is the strongest decoder by exact correction-label accuracy and component accuracy. The MLP ties the lookup baseline, while the QNN trails the baseline slightly and is substantially more expensive to train in the reported simulation. A fair next step is to give all learned models equivalent input information, repeat experiments over multiple random seeds, report uncertainty, and implement logical-observable evaluation before drawing claims about actual error-correction performance.